

Predictive Analytics Using Historical Data

Data Analytics Internship Project

Objective

To build a predictive model using historical data and forecast future trends using machine learning techniques.

Abstract

Predictive analytics uses historical data, statistical algorithms, and machine learning techniques to predict future outcomes. This project demonstrates trend forecasting using a simple regression model.

Tools Used

Python, Pandas, NumPy, Scikit-Learn, Matplotlib

Sample Historical Dataset

Monthly sales data from previous years was used as sample historical data for forecasting.

Methodology

1. Data Collection
2. Data Cleaning and Preprocessing
3. Feature Selection
4. Linear Regression Model Training
5. Forecast Generation
6. Accuracy Evaluation and Visualization

Results

The predictive model successfully identified trends in historical data and generated future forecasts. The model can assist organizations in planning and decision-making.

Conclusion

Predictive analytics enables data-driven forecasting and improves business planning. Historical data can be effectively used to predict future trends using regression techniques.